Lesson notes · for you

Measure it right

IoT · Level 1 · Lesson 3

Materials
A ruler or tape measure (a tailor's tape works); Paper and a pencil for your table; Optional: a thermometer, or a bottle rain gauge (an adult cuts the bottle)

Learning goals

- Read a scale carefully: the unit, eye level and what each small step is worth.
- Record a measurement with its number, unit, time and place.
- Spot a reading that can't be right, and say what to do about it.

Activity: Measure it three times

1. Pick something you can measure several times: your desk with a ruler or tape, the shade temperature with a thermometer, or rain in a bottle gauge.
2. Take two numbered marks, subtract, and divide by the gaps between them. Write the answer at the top of your page.
3. Bring your eye level with the mark each time. Use the same tool, the same place and the same method.
4. Make a table with columns for number, unit, time and place, and fill in one row for each reading.
5. Ask a friend to measure the same thing. If you disagree, look for the reason (the unit, eye level, the place) before changing anything.

If you make a bottle rain gauge, an adult cuts the bottle and tapes the sharp edge. If a glass thermometer breaks, don't touch it: tell an adult.

Discussion

- Why might two careful people get slightly different readings?
- When is it fair to leave a reading out, and what must you write down?
- Where in daily life do you see numbers with no unit?

Check yourself

- There are 5 gaps between 0 and 10 on a scale. What is each gap worth?

Try it this week
Measure your desk three times, the same way, and write each reading with its number, unit, time and place.

Worksheet

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Name _____ Class _____ Date _____

1 **The school garden's soil moisture sensor reads 50 %, then exactly 0 % at midnight, then 50 % again an hour later. Nobody watered. What should you do with the 0?**

- ◯ Delete it, so the chart looks clean and nobody is confused
- ◯ Keep it, mark it as a glitch, and check the sensor's wires
- ◯ Water the garden at once: the sensor says it's bone dry

2 **Read Yusuf's rain gauge.**

The marks 0 and 20 mm have 4 small gaps between them, so each gap is worth _____ mm. The water stands 3 gaps above 0, so _____ mm of rain fell.

Word bank: 15 · 12 · 5 · 3

3 **You measure your desk three times and get 70, 71 and 70 cm. That means you made a mistake.**

◯ True ◯ False

4 **Put the steps of keeping a good temperature log in order.**

- ☐ Look for readings that can't be right
- ☐ Mark any glitch, keep it, and note why you left it out
- ☐ Read the thermometer with your eye level with the mark
- ☐ Choose one shady place and one time each day
- ☐ Write the number, unit, time and place

5 **Which of these records could someone compare with next week's?**

☐ 31 °C, 2 pm Tuesday, back of the classroom ☐ 31 °C, back of the classroom

☐ Rain: 12 mm since 8 am Monday, school garden gauge ☐ Very hot, after lunch ☐ 31

✎ **Think of a time a number without a unit, or a wrong reading, confused someone: a recipe, a price, a test score. What would have made it clear?**

Check your answers

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1 **Keep it, mark it as a glitch, and check the sensor's wires**

Soil can't dry out and get wet again in an hour with nobody watering, so the 0 is a glitch, often a loose wire.

2 **The marks 0 and 20 mm have 4 small gaps between them, so each gap is worth 5 mm. The water stands 3 gaps above 0, so 15 mm of rain fell.**

20 ÷ 4 = 5 mm a gap, and 3 gaps × 5 mm = 15 mm.

3 **False**

Every measurement differs a little, even when you're careful.

4 **Top to bottom: 4, 5, 2, 1, 3**

Plan first, so every reading is taken the same way; read carefully; record it fully; and only then check the log and handle glitches openly.

5 **31 °C, 2 pm Tuesday, back of the classroom; Rain: 12 mm since 8 am Monday, school garden gauge**

A useful record has the number, unit, time and place.